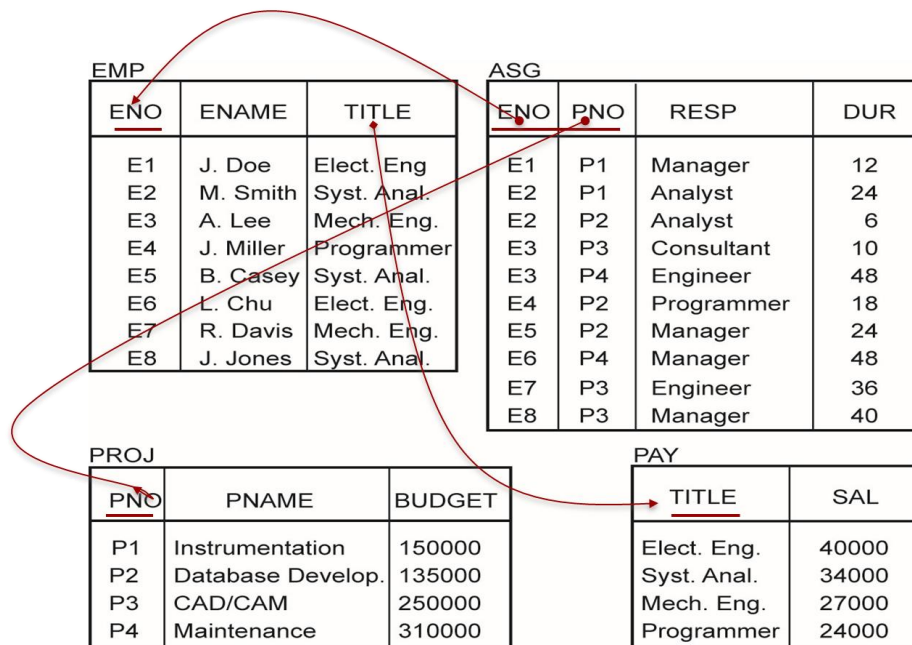


Tutorial 2: Distributed Query Processing

Question 1: At the global level, distributed query processing consists of three main steps: query decomposition, data localization and global optimization. Discuss these three steps, focusing on the input, output, and objectives for each step.

Question 2: Consider the following database (arrows are foreign-key linkages):



(a) Assume that relation PROJ is horizontally fragmented as follows:

$$PROJ_1 = \sigma_{PNO \leq "P2"} PROJ$$

$$PROJ_2 = \sigma_{PNO > "P2"} PROJ$$

Transform the following query into a reduced query on fragments:

```
SELECT ENO, PNAMESEP
FROM PROJ, ASGSEP
WHERE PROJ.PNO = ASG.PNO AND PNO = "P4"
```

(b) Assume PROJ is fragmented as above, and ASG is fragmented as below:

$$ASG_1 = \sigma_{PNO \leq "P2"} ASG$$

$$ASG_2 = \sigma_{"P2" < PNO \leq "P3"} ASG$$

$$ASG_3 = \sigma_{PNO > "P3"} ASG_{SEP}^{[L]}$$

Transform the following query into a reduced query on fragments, and determine whether it is better than the localized query: $_{SEP}^{[L]}$

```
SELECT RESP, BUDGET
FROM ASG, PROJ $_{SEP}^{[L]}$ 
WHERE ASG.PNO = PROJ.PNO AND PNAME = "CAD/CAM"
```

Question 3: Let R(A, B) and S(B, C, D) be two relations as shown below:

R(A, B)

A	B
1	4
1	5
2	4
2	6
3	7

S(B, C, D)

B	C	D
4	5	0
4	7	8
5	0	1
5	1	1

- Compute $R \bowtie S$
- Compute $S \bowtie R$
- Assume R is at site 1 and S is at site 2, and a query $R \bowtie S$ is issued at site 2. List the steps for a query processing strategy using semi-join, and check if the semi-join is a beneficial option in this case (ignore local processing cost).

---ooo000000000ooo---